

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Tuesday, 18 August 2026

Contents

1	Integrals	2
1.1	Medium Integral	2
1.2	Hard Integral	2
2	Differentials	4
2.1	Medium Differential	4
2.2	Hard Differential	4
3	Derivatives	6
3.1	Medium Derivative	6
3.2	Hard Derivative	6
4	Physics & Engineering	8
4.1	Mechanics	8
4.2	Quantum Mechanics	8
4.3	Electric Fields	9

Integrals

1.1 Medium Integral

Definite Integral with Trigonometry

Evaluate the definite integral:

$$\int_0^{\pi} x \sin^3(x) dx$$

Solution: Let I be the integral. We employ the property of definite integrals:

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

Applying this to our integral, we replace x with $(\pi - x)$:

$$I = \int_0^{\pi} (\pi - x) \sin^3(\pi - x) dx$$

Since $\sin(\pi - x) = \sin(x)$, the integral becomes:

$$I = \int_0^{\pi} (\pi - x) \sin^3(x) dx = \pi \int_0^{\pi} \sin^3(x) dx - \int_0^{\pi} x \sin^3(x) dx$$

Notice that the second term on the right is exactly our original integral I . Adding I to both sides yields:

$$2I = \pi \int_0^{\pi} \sin^3(x) dx \implies I = \frac{\pi}{2} \int_0^{\pi} \sin^3(x) dx$$

Now we evaluate the remaining integral by writing $\sin^3(x) = \sin(x) \sin^2(x) = \sin(x)(1 - \cos^2(x))$:

$$I = \frac{\pi}{2} \int_0^{\pi} (1 - \cos^2(x)) \sin(x) dx$$

Let $u = \cos(x)$. Then $du = -\sin(x) dx$. The limits of integration change from $x = 0 \implies u = 1$ to $x = \pi \implies u = -1$.

$$I = \frac{\pi}{2} \int_1^{-1} -(1 - u^2) du = \frac{\pi}{2} \int_{-1}^1 (1 - u^2) du$$

Because $(1 - u^2)$ is an even function:

$$I = \pi \int_0^1 (1 - u^2) du = \pi \left[u - \frac{u^3}{3} \right]_0^1 = \pi \left(1 - \frac{1}{3} \right) = \pi \left(\frac{2}{3} \right)$$

The exact value is $\frac{2\pi}{3}$.

1.2 Hard Integral

Multiple Integral of a Minimum Function

Evaluate the multiple definite integral:

$$\int_0^1 \int_0^1 \cdots \int_0^1 \min(x_1, x_2, \dots, x_{22}) dx_1 dx_2 \dots dx_{22}$$

Solution: Let $n = 22$. We can interpret this integral probabilistically. The integral represents the expected value of the minimum of n independent random variables $X_1, X_2, \dots, X_n$, each uniformly distributed on the interval $[0, 1]$. Let $M = \min(X_1, X_2, \dots, X_n)$. To find the expected value $E[M]$, we first determine its cumulative distribution function. For M to be greater than x , all individual variables X_i must be greater than x :

$$P(M > x) = P(X_1 > x, X_2 > x, \dots, X_n > x)$$

Since the variables are independent and uniform on $[0, 1]$:

$$P(X_i > x) = 1 - x \text{ for } x \in [0, 1]$$

Thus:

$$P(M > x) = (1 - x)^n$$

For a non-negative random variable bounded by 1, the expected value can be computed directly by integrating the survival function $P(M > x)$:

$$E[M] = \int_0^1 P(M > x) dx = \int_0^1 (1 - x)^n dx$$

Let $u = 1 - x$, then $du = -dx$. The bounds reverse:

$$E[M] = \int_1^0 -u^n du = \int_0^1 u^n du = \left[\frac{u^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1}$$

For our specific problem, $n = 22$. Substituting this in:

$$\int_0^1 \int_0^1 \cdots \int_0^1 \min(x_1, x_2, \dots, x_{22}) dx_1 \dots dx_{22} = \frac{1}{22+1} = \frac{1}{23}$$

Differentials

2.1 Medium Differential

First-Order Linear Initial Value Problem

Find the specific value of $y(1)$ given the first-order linear initial value problem:

$$\frac{dy}{dx} - \frac{2x}{x^2 + 1}y = (x^2 + 1)e^x$$

With the initial condition $y(0) = 2$.

Solution: The given differential equation is a standard first-order linear ordinary differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$. Here, $P(x) = -\frac{2x}{x^2+1}$. We compute the integrating factor, $\mu(x)$:

$$\mu(x) = e^{\int P(x) dx} = \exp\left(\int -\frac{2x}{x^2 + 1} dx\right)$$

Using the substitution $u = x^2 + 1$, $du = 2x dx$:

$$\mu(x) = \exp(-\ln(x^2 + 1)) = \frac{1}{x^2 + 1}$$

Multiplying the entire differential equation by the integrating factor gives:

$$\frac{1}{x^2 + 1} \frac{dy}{dx} - \frac{2x}{(x^2 + 1)^2}y = e^x$$

By the product rule, the left side simplifies to the derivative of $(y \cdot \mu(x))$:

$$\frac{d}{dx} \left(y \cdot \frac{1}{x^2 + 1} \right) = e^x$$

Integrating both sides with respect to x :

$$\frac{y}{x^2 + 1} = \int e^x dx \implies \frac{y}{x^2 + 1} = e^x + C$$

$$y(x) = (x^2 + 1)(e^x + C)$$

We use the initial condition $y(0) = 2$ to solve for C :

$$2 = (0^2 + 1)(e^0 + C) \implies 2 = 1(1 + C) \implies C = 1$$

The specific solution to the differential equation is:

$$y(x) = (x^2 + 1)(e^x + 1)$$

To find $y(1)$, we substitute $x = 1$:

$$y(1) = (1^2 + 1)(e^1 + 1) = 2(e + 1) = 2e + 2$$

2.2 Hard Differential

Exact Equations & Differentials

Solve the differential equation:

$$\frac{dy}{dx} = \frac{-\cos(x)(3\cos(y) - 7\sin(x) - 3)}{\sin(y)(3\sin(x) - 7\cos(y) + 7)}$$

Express the solution in the form:

$$(\cos(y) - \sin(x) - 1)^2(\sin(x) + \cos(y) - 1)^n = C$$

Find the value of n .

Solution: Let $u = \sin(x)$ and $v = \cos(y)$. Differentiating gives $du = \cos(x) dx$ and $dv = -\sin(y) dy$. Substituting these into the original ODE:

$$\frac{-dv}{du} = \frac{-(3v - 7u - 3)}{3u - 7v + 7} \implies \frac{dv}{du} = \frac{3v - 7u - 3}{3u - 7v + 7}$$

Rearranging into standard differential form $M du + N dv = 0$:

$$(3v - 7u - 3) du + (7v - 3u - 7) dv = 0$$

We are given that the solution is of the form $(v - u - 1)^2(u + v - 1)^n = C$. Let's take the natural logarithm and implicitly differentiate:

$$2\ln(v - u - 1) + n\ln(u + v - 1) = \ln(C)$$

$$\frac{2(dv - du)}{v - u - 1} + \frac{n(du + dv)}{u + v - 1} = 0$$

Multiplying by the common denominator $(v - u - 1)(u + v - 1)$:

$$2(u + v - 1)(dv - du) + n(v - u - 1)(du + dv) = 0$$

Expanding and collecting the du and dv terms:

$$[-2(u + v - 1) + n(v - u - 1)] du + [2(u + v - 1) + n(v - u - 1)] dv = 0$$

Grouping by u , v , and constants:

$$[-(n + 2)u + (n - 2)v + (2 - n)] du + [(2 - n)u + (n + 2)v - (n + 2)] dv = 0$$

For this to match the derived differential equation $(3v - 7u - 3) du + (7v - 3u - 7) dv = 0$, the coefficients must be strictly proportional. Comparing the dv coefficients:

$$(2 - n)u + (n + 2)v - (n + 2) \equiv k(-3u + 7v - 7)$$

Equating coefficients implies $k = 1$ and:

$$2 - n = -3 \implies n = 5$$

Checking this against the v term: $n + 2 = 7 \implies 5 + 2 = 7$, which matches perfectly. Checking the du term with $n = 5$:

$$-(5 + 2)u + (5 - 2)v + (2 - 5) = -7u + 3v - 3 = 3v - 7u - 3$$

This perfectly matches the du coefficient of our transformed ODE. Thus, $n = 5$.

Derivatives

3.1 Medium Derivative

Logarithmic Exponentiation

Let $f(x) = \log_x(\ln(x^x))$. Find $f'(e)$.

Solution: First, we simplify the function $f(x)$ using logarithm properties. We know that $\ln(x^x) = x \ln(x)$. Substituting this into the function:

$$f(x) = \log_x(x \ln(x))$$

Using the change of base formula, $\log_a(b) = \frac{\ln(b)}{\ln(a)}$:

$$f(x) = \frac{\ln(x \ln(x))}{\ln(x)}$$

Using the product rule for logarithms on the numerator, $\ln(ab) = \ln(a) + \ln(b)$:

$$f(x) = \frac{\ln(x) + \ln(\ln(x))}{\ln(x)} = 1 + \frac{\ln(\ln(x))}{\ln(x)}$$

Now we find the derivative $f'(x)$ using the quotient rule for the second term:

$$f'(x) = 0 + \frac{\frac{d}{dx}[\ln(\ln(x))] \cdot \ln(x) - \ln(\ln(x)) \cdot \frac{d}{dx}[\ln(x)]}{(\ln(x))^2}$$

The derivative of $\ln(\ln(x))$ is $\frac{1}{x \ln(x)}$ via the chain rule.

$$f'(x) = \frac{\left(\frac{1}{x \ln(x)}\right) \ln(x) - \ln(\ln(x)) \left(\frac{1}{x}\right)}{(\ln(x))^2}$$

$$f'(x) = \frac{\frac{1}{x} - \frac{\ln(\ln(x))}{x}}{(\ln(x))^2} = \frac{1 - \ln(\ln(x))}{x(\ln(x))^2}$$

Now we evaluate this derivative at $x = e$:

$$f'(e) = \frac{1 - \ln(\ln(e))}{e(\ln(e))^2}$$

Since $\ln(e) = 1$, we have $\ln(\ln(e)) = \ln(1) = 0$:

$$f'(e) = \frac{1 - 0}{e(1)^2} = \frac{1}{e}$$

3.2 Hard Derivative

Higher Order Trigonometric Derivative

Let $f(x) = \sin(x) \cos(2x)$. Solve for the integer x if:

$$f^{(5)}(0) = x^4 + x^3 + x^2 + x + 1$$

Solution: First, we rewrite $f(x)$ using the trigonometric product-to-sum identities to make differentiation easier:

$$\sin(A) \cos(B) = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$f(x) = \frac{1}{2} [\sin(x + 2x) + \sin(x - 2x)] = \frac{1}{2} [\sin(3x) + \sin(-x)]$$

Since sine is an odd function, $\sin(-x) = -\sin(x)$:

$$f(x) = \frac{1}{2} \sin(3x) - \frac{1}{2} \sin(x)$$

We need to find the fifth derivative evaluated at 0. The n -th derivative of $\sin(kx)$ is $k^n \sin(kx + \frac{n\pi}{2})$. For $n = 5$:

$$\frac{d^5}{dx^5} \sin(kx) = k^5 \sin\left(kx + \frac{5\pi}{2}\right) = k^5 \cos(kx)$$

Therefore, the fifth derivative of our function is:

$$f^{(5)}(x) = \frac{1}{2}(3^5) \cos(3x) - \frac{1}{2}(1^5) \cos(x)$$

Evaluating at $x = 0$, where $\cos(0) = 1$:

$$f^{(5)}(0) = \frac{1}{2}(243)(1) - \frac{1}{2}(1)(1) = \frac{243 - 1}{2} = \frac{242}{2} = 121$$

We are given that $f^{(5)}(0) = x^4 + x^3 + x^2 + x + 1$. Equating the two values:

$$x^4 + x^3 + x^2 + x + 1 = 121 \implies x^4 + x^3 + x^2 + x - 120 = 0$$

We check small integer values for x : If $x = 1$: $1 + 1 + 1 + 1 - 120 \neq 0$ If $x = 2$: $16 + 8 + 4 + 2 - 120 \neq 0$ If $x = 3$: $81 + 27 + 9 + 3 - 120 = 120 - 120 = 0$ Thus, the integer solution is $x = 3$.

Physics & Engineering

4.1 Mechanics

U-Tube Oscillations

A U-tube has uniform cross-sectional area $A = 3.0 \times 10^{-4} \text{ m}^2$. It contains an incompressible liquid of density $\rho = 850 \text{ kg m}^{-3}$. The total length of the moving liquid is $L = 1.20 \text{ m}$. One surface is displaced upward by a small distance x . The other surface moves downward by the same distance. Both free surfaces are exposed to the same external pressure. Neglect resistance and find the period T in seconds. Take $g = 9.81 \text{ m s}^{-2}$.

Solution: When the liquid is displaced by x on one side, it drops by x on the other side. This creates a difference in liquid height of $2x$ between the two columns. The restoring force F is equal to the weight of this excess liquid column:

$$F = -mg = -(\rho V)g = -(\rho A(2x))g = -2\rho Agx$$

Using Newton's Second Law, $F = m_{\text{total}}a$. The total mass of the liquid moving is $m_{\text{total}} = \rho AL$, and acceleration is $a = \frac{d^2x}{dt^2}$.

$$(\rho AL) \frac{d^2x}{dt^2} = -2\rho Agx$$

Dividing both sides by ρA :

$$L \frac{d^2x}{dt^2} + 2gx = 0 \implies \frac{d^2x}{dt^2} + \left(\frac{2g}{L}\right)x = 0$$

This is the differential equation for simple harmonic motion (SHM) of the form $\frac{d^2x}{dt^2} + \omega^2x = 0$, where $\omega = \sqrt{\frac{2g}{L}}$. The period T of oscillation is given by $T = \frac{2\pi}{\omega}$:

$$T = 2\pi \sqrt{\frac{L}{2g}}$$

Substitute the given values ($L = 1.20$, $g = 9.81$):

$$T = 2\pi \sqrt{\frac{1.20}{2(9.81)}} = 2\pi \sqrt{\frac{1.20}{19.62}} \approx 2\pi \sqrt{0.061162} \approx 2\pi(0.2473) \approx 1.55 \text{ s}$$

The period is approximately 1.55 s.

4.2 Quantum Mechanics

Compton Wavelength & X-Rays

An electron is accelerated from rest until its de Broglie wavelength is three times its Compton wavelength, $\lambda = 3\lambda_C$, where $\lambda_C = \frac{h}{m_e c}$. It then strikes a target and produces X-rays. Assume nonrelativistic motion and complete conversion of its kinetic energy into the energy of one photon. Find the coefficient k if the minimum X-ray wavelength satisfies $\lambda_{\min} = k\lambda_C$.

Solution: The de Broglie wavelength of the electron is $\lambda = \frac{h}{p}$. We are given that $\lambda = 3\lambda_C$.

$$\frac{h}{p} = 3 \left(\frac{h}{m_e c} \right) \Rightarrow p = \frac{m_e c}{3}$$

Assuming non-relativistic motion, the kinetic energy K of the electron is:

$$K = \frac{p^2}{2m_e} = \frac{\left(\frac{m_e c}{3}\right)^2}{2m_e} = \frac{m_e^2 c^2 / 9}{2m_e} = \frac{m_e c^2}{18}$$

Upon striking the target, this kinetic energy is entirely converted into the energy of a single photon. The energy of this photon is $E = \frac{hc}{\lambda_{\min}}$. Equating the energies:

$$\frac{hc}{\lambda_{\min}} = K = \frac{m_e c^2}{18}$$

Solving for the minimum X-ray wavelength $\lambda_{\min}$:

$$\lambda_{\min} = \frac{18hc}{m_e c^2} = 18 \left(\frac{h}{m_e c} \right) = 18\lambda_C$$

Since $\lambda_{\min} = k\lambda_C$, we match the coefficients directly to find $k = 18$.

4.3 Electric Fields

Stretching a Wire

A cylindrical wire has an initial resistance of 18.00Ω . It is stretched uniformly until $L_2 = 1.30L_1$. Its volume and resistivity remain constant. Find its final resistance in ohms. Give your answer to two decimal places.

Solution: The resistance of a cylindrical wire is given by $R = \rho \frac{L}{A}$. For the initial state: $R_1 = \rho \frac{L_1}{A_1} = 18.00 \Omega$. Since the volume V remains constant during stretching, $V = L_1 A_1 = L_2 A_2$. Therefore, the new cross-sectional area is:

$$A_2 = \frac{L_1}{L_2} A_1$$

The final resistance R_2 is:

$$R_2 = \rho \frac{L_2}{A_2} = \rho \frac{L_2}{\left(\frac{L_1}{L_2} A_1\right)} = \rho \frac{L_2^2}{L_1 A_1} = \left(\frac{L_2}{L_1}\right)^2 \left(\rho \frac{L_1}{A_1}\right) = \left(\frac{L_2}{L_1}\right)^2 R_1$$

We are given that the wire is stretched such that $L_2 = 1.30L_1$, meaning the ratio $\frac{L_2}{L_1} = 1.30$. Substitute the values into the resistance equation:

$$R_2 = (1.30)^2 \times 18.00 = 1.69 \times 18.00 = 30.42 \, \Omega$$

The final resistance is exactly $30.42 \, \Omega$.